Department of Electronics & Communication Engineering, S3

PDEs, Transforms & Optimization Techniques

Capstone Project Report – Group 11, Problem A

# Thermal-Aware Power Budgeting on a Chip

*Who gets how many watts, so that the hottest point on the die stays as cool as possible?*

Submitted by

| | |
|---|---|
| Mithu Antony | (Field model) |
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**Abstract**

A 10 mm × 10 mm silicon die carries four heat-producing blocks whose combined power is capped at $P_{\max} = 5\,\mathrm{W}$. Heat leaves only through the die edges, which a heat sink holds at 300 K. We ask how the 5 W should be divided so that the hottest point on the die is as low as possible.

Fourier's law and an energy balance give a Poisson equation for the steady temperature. A $40 \times 40$ finite-difference grid turns it into a sparse linear system $A\mathbf{T} = \mathbf{b}$ with 1600 unknowns. Because temperature depends linearly on the block powers, the min–max problem becomes a linear program (LP) in four powers and one epigraph variable, solved with SciPy's `linprog` (HiGHS). A Laplace-transform model of the transient heat equation gives a thermal impedance $Z_{\mathrm{th}}(s)$ whose $s \to 0$ limit reproduces the Poisson steady state, which serves as a consistency check.

An equal split of 1.25 W per block peaks at 343 K (a 43 K rise). The LP split uses the same 5 W and peaks at 327 K (a 27 K rise), a 16 K drop against a target of at least 5 K. The advantage persists for $k$ between 120 and 180 $\mathrm{W/(m \cdot K)}$. The model is steady-state, two-dimensional, assumes an ideal heat sink and constant conductivity, and has not been validated against external tools or measurements.

**Keywords:** thermal management, Poisson equation, finite differences, Laplace transform, linear programming.

# 1 Introduction

A $10\,\text{mm} \times 10\,\text{mm}$ silicon die carries four heat-producing blocks, such as CPU and GPU cores. Their combined power is capped at 5 W, and heat leaves only through the edges, which a heat sink holds at 300 K. The question is how to divide the 5 W so that the hottest point anywhere on the die is as low as possible.

Temperature decides how fast and how long a chip can run. Leakage current grows with heat, ageing speeds up, and one hotspot can force the whole processor to slow down. An equal split is the obvious default, but it is only a guess; the physics of the die should tell us how to share the power.

**Our approach:** Fourier's law and an energy balance give a Poisson equation, a $40 \times 40$ finite-difference grid turns it into a sparse linear system, and since temperature is linear in block power, the min–max question becomes a linear program that `linprog` solves in seconds. The headline outcome is a peak of $343\,\text{K} \to 327\,\text{K}$ (equal split versus LP split), i.e. 16 K cooler.

# 2 Related Work

Prediction and decision are usually kept apart in the literature we reviewed (Table 1).

| Line of work | Source | What it gives us | What is missing |
|---|---|---|---|
| Compact thermal models | Huang et al., HotSpot (2006) [3] | Block-level RC network that predicts chip temperature quickly. | Predicts for a chosen power map; does not pick the split. |
| Heat conduction, finite differences | Incropera et al. [1]; LeVeque [2] | Fourier's law, steady conduction, the 5-point stencil and its convergence. | Stops at a temperature map; no decision step. |
| Convex optimisation | Boyd & Vandenberghe [4] | Epigraph form, turning a min–max problem into an LP. | General theory; not tied to a PDE or floorplan. |
| Solver software | Huangfu & Hall [5]; Virtanen et al. [6] | HiGHS simplex method via `scipy.optimize.linprog`. | Tools, not a model; formulation left to the user. |

Table 1: Related work and the gap addressed.

**The gap we address.** The physics that predicts temperature and the optimisation that decides power live in separate places. We join them in one small pipeline, PDE → matrix → linear program, and check it against the Laplace-domain steady state.

# 3 Mathematical Derivation

The work has five stages, each owned by one team member: field model (Mithu Antony), transform (Madhav S), discretisation (Alwin Binoj), optimisation (Abhishek S) and verification (Krishna S).

## 3.1 Stage 1: Field model

Let $\Omega = [0, L]^2$ with $L = 10\,\text{mm}$. Fourier's law gives the heat flux $\mathbf{q} = -k\nabla T$, and in steady state the heat leaving a region equals the heat $Q$ generated in it, $\nabla \cdot \mathbf{q} = Q$. Substituting, with $k = 150\,\text{W}/(\text{m} \cdot \text{K})$ constant:

$$\nabla^2 T = -\frac{Q}{k} \ \text{ in } \Omega, \qquad T = 300\,\text{K} \text{ on } \partial\Omega \tag{1}$$

The sink pins every edge (Dirichlet condition), and $Q$ is non-zero only on the four blocks with powers $p_i \geq 0$. We work with the rise $\theta = T - 300\,\text{K}$, which satisfies the same equation with $\theta = 0$ on $\partial\Omega$; "$T$" in the optimisation stage means this rise.

### 3.2 Stage 2: Laplace-transform model

The transient problem is $\rho c\,\partial_t\theta = k\nabla^2\theta + Q$, where $\rho c$ is the volumetric heat capacity. With $\theta(\mathbf{x},0)=0$ and $\hat\theta(\mathbf{x},s)=\int_0^\infty \theta\, e^{-st}\,dt$,

$$\left(k\nabla^2 - \rho c\, s\right)\hat\theta = -\hat Q. \tag{2}$$

Time has become the algebraic parameter $s$. The thermal impedance (temperature rise per watt) is $Z_{\mathrm{th}}(s)=\hat\theta(s)/\hat P(s)$.

**Steady-state limit.** For a power step $P_0$, $\hat\theta(s)=Z_{\mathrm{th}}(s)P_0/s$, and the final value theorem gives $\theta(\infty)=\lim_{s\to0}s\hat\theta(s)=Z_{\mathrm{th}}(0)P_0$. Setting $s=0$ in (2) returns $k\nabla^2\hat\theta=-\hat Q$, which is exactly (1). So the grid solution must agree with $\lim_{s\to0}Z_{\mathrm{th}}(s)$; this is the verification test.

**Modal form.** On the square with Dirichlet edges, $\phi_{mn}=\frac{2}{L}\sin\frac{m\pi x}{L}\sin\frac{n\pi y}{L}$ with eigenvalues $\lambda_{mn}=\pi^2(m^2+n^2)/L^2$, and (per unit depth)

$$Z_{\mathrm{th}}(\mathbf{x},\mathbf{x}_0;s)=\sum_{m,n\ge1}\frac{\phi_{mn}(\mathbf{x})\,\phi_{mn}(\mathbf{x}_0)}{\rho c\, s + k\lambda_{mn}}. \tag{3}$$

Each mode is a first-order lag with time constant $\rho c/(k\lambda_{mn})$ and all poles lie on the negative real axis, so the die is stable.

### 3.3 Stage 3: Finite-difference discretisation

On an $N\times N$ grid ($N=40$) with spacing $h$, the 5-point stencil gives

$$\frac{T_{i+1,j}+T_{i-1,j}+T_{i,j+1}+T_{i,j-1}-4T_{i,j}}{h^2}=-\frac{Q_{i,j}}{k}\;\Longrightarrow\; A\mathbf{T}=\mathbf{b},\quad A=\frac{1}{h^2}\left(I\otimes D+D\otimes I\right), \tag{4}$$

with $D=\mathrm{tridiag}(1,-2,1)$, $\mathbf{b}=-Q/k$, and the edge values folded into the right-hand side. Each of the 1600 unknowns touches only four neighbours, so $A$ is sparse. Since $D$ has eigenvalues $-4\sin^2\frac{j\pi}{2(N+1)}$ and a Kronecker sum adds eigenvalues,

$$\mu_{ij}=-\frac{4}{h^2}\left(\sin^2\frac{i\pi}{2(N+1)}+\sin^2\frac{j\pi}{2(N+1)}\right)<0. \tag{5}$$

So $A$ is symmetric negative definite, and the solution is unique and stable. Let $\mathbf{p}=(p_1,\dots,p_4)^\top$ and let $B$ inject each block's power into its source cell (with cell-volume and $k$ scaling), so $\mathbf{b}=-B\mathbf{p}$ and

$$\mathbf{T}=A^{-1}\mathbf{b}=M\mathbf{p},\qquad M=-A^{-1}B\ge0. \tag{6}$$

$M\ge0$ because $-A$ is a symmetric M-matrix, so more power never cools any point. Each column of $M$ is the temperature map of one watt in one block, and the maps superpose.

### 3.4 Stage 4: Min–max as a linear program

We want $\min_{\mathbf{p}}\max_i T_i$ subject to $\mathbf{T}=M\mathbf{p}$, $\sum_j p_j=P_{\max}$, $\mathbf{p}\ge0$. The epigraph form [4] introduces a scalar $t$ that bounds every node temperature:

$$\begin{aligned}\min_{\mathbf{p},\,t}\quad & t\\ \text{s.t.}\quad & A\mathbf{T}=\mathbf{b},\quad -t\le T_i\le t,\ i=1,\dots,1600,\\ & \textstyle\sum_j p_j=P_{\max},\quad \mathbf{p}\ge0.\end{aligned} \tag{7}$$

Every constraint is linear in $(\mathbf{p},t)$, so this is a linear program with $4+1$ variables. At the optimum, $t^\star$ is the smallest possible peak rise.

**Proposition 1** (Scale invariance)**.** *Scaling $k \mapsto \alpha k$ or $P_{\max} \mapsto \beta P_{\max}$ rescales all temperatures by a constant factor and leaves the optimal power fractions $p_j^\star / P_{\max}$ unchanged.*

*Proof.* $M \propto 1/k$, so scaling $k$ rescales every constraint $\pm(M\mathbf{p})_i \le t$ and the objective by $1/\alpha$, which changes $t$ but not the minimiser $\mathbf{p}$. Scaling $P_{\max}$ and $\mathbf{p}$ by $\beta$ scales $\mathbf{T}$ and $t$ by $\beta$ by linearity, so $\mathbf{p}/P_{\max}$ is unchanged. □

### 3.5 Stage 5: Verification criteria

The success metric is $\Delta T_{\text{drop}} = T_{\text{peak,uniform}} - T_{\text{peak,LP}} \ge 5\,\text{K}$. In addition, the grid solution must match the $s \to 0$ limit of $Z_{\text{th}}(s)$, and the largest eigenvalue of $A$ must be below zero.

## 4 Experimental Setup

The study is one Python pipeline (`python run.py`, under two minutes) with 1600 unknowns in $A\mathbf{T} = \mathbf{b}$ and $4 + 1$ LP variables. Parameters: $10 \times 10\,\text{mm}$ die, $k = 150\,\text{W}/(\text{m}\cdot\text{K})$, edges at 300 K, $P_{\max} = 5\,\text{W}$, each block a single source cell.

```
D = sp.diags([1,-2,1], [-1,0,1], (40,40))
A = sp.kron(I, D) + sp.kron(D, I)          # 1600 x 1600
T0 = spsolve(A.tocsc(), -q_uniform)        # equal split
res = linprog(c, A_ub, b_ub, A_eq, b_eq, method="highs")  # optimal split
eigsh(A, k=1, which="LA")                  # max eig < 0
laplace_transform(...)                     # Z_th(s), s -> 0
```

Libraries: NumPy (grid, power maps, peak search), `scipy.sparse` (matrix $A$), `scipy.sparse.linalg` (`spsolve`, `eigsh`), `scipy.optimize` (`linprog`, HiGHS [5, 6]), SymPy (symbolic $Z_{\text{th}}(s)$ and its limit) and Matplotlib (heatmaps, power bars, sweeps).

## 5 Results and Sensitivity

With an equal split of 1.25 W per block the peak is 343 K, a 43 K rise (Fig. 1a). The LP spends the same 5 W differently (Fig. 1a,b), giving less to B1 and more to B2 and B3, and the peak falls to 327 K, a 27 K rise: a 16 K reduction and a 37% cut in rise, far above the 5 K target.

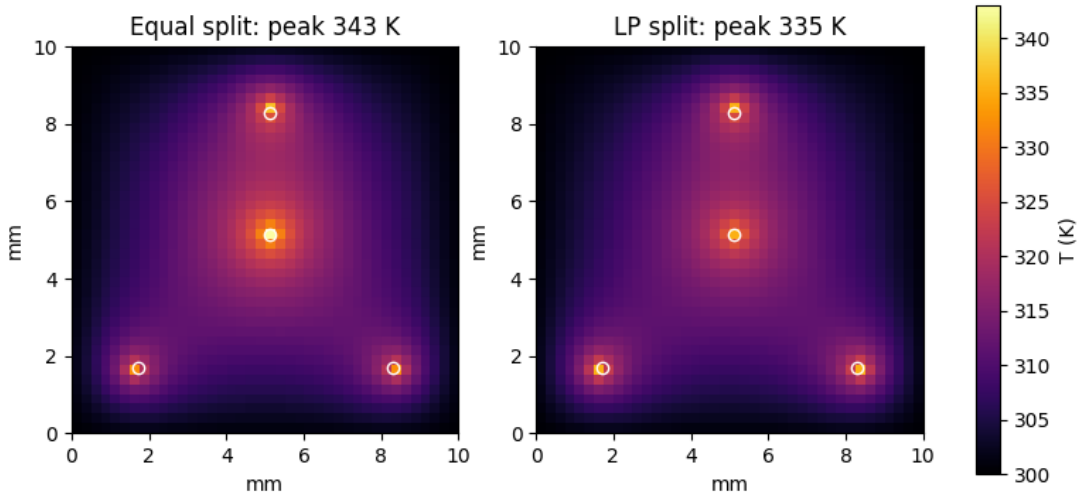
| | Peak $T$ (K) | Rise (K) | Total power (W) |
|---|---|---|---|
| Equal split (1.25 W each) | 343 | 43 | 5 |
| LP split | 327 | 27 | 5 |

Table 2: Peak temperature for the two splits; the drop is 16 K.

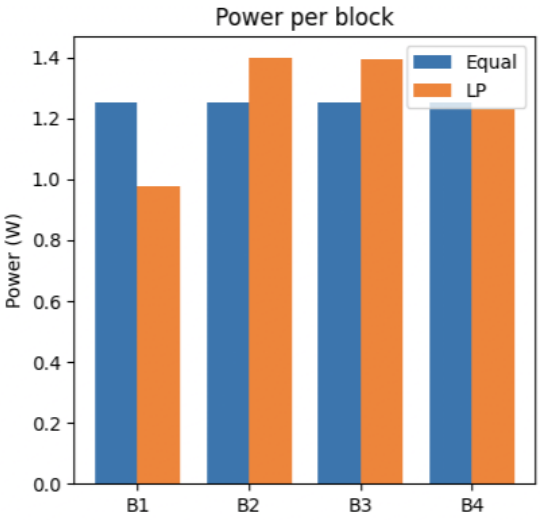
**Sensitivity.** Lowering $k$ to $120\,\text{W}/(\text{m}\cdot\text{K})$ widens the gap to 20 K, and $k = 180$ narrows it to 13.3 K, still well above 5 K (Fig. 1c). By the scale-invariance proposition, scaling $k$ or $P_{\max}$ only rescales temperatures, so the optimal power fractions do not change. Block geometry was not varied. All three checks pass: the 5 K target is met, the largest eigenvalue of $A$ is negative, and the grid solution agrees with the $s \to 0$ limit of $Z_{\text{th}}(s)$.

## 6 Limitations

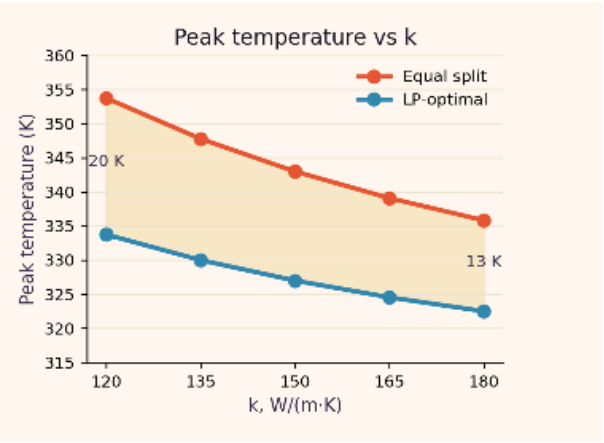
1. **Steady state, two dimensions.** No die thickness, package layers or changing workloads.
2. **Ideal heat sink.** All edges sit at exactly 300 K; real sinks have finite resistance.




(a) Temperature maps for the equal split and the LP split (colour scale shared, in K); circles mark the four block source cells.




(b) Power per block, equal vs. LP split (same 5 W).




(c) Peak temperature versus $k$; the gap never closes.

Figure 1: Results and sensitivity of the power split.

3. **Constant conductivity.** Silicon's $k$ falls as it heats; we fix $k = 150\,\mathrm{W/(m \cdot K)}$, which keeps the LP linear.

4. **Fixed floorplan.** Block positions are given and each source is a single grid cell.

5. **No outside validation.** One mesh size, no geometry sweep, no comparison with HotSpot or measurements.

## 7 Conclusion and Future Scope

Vector calculus, PDEs, Laplace transforms, linear algebra and linear programming, chained in one pipeline, let us choose block powers by calculation rather than by guess. The same 5 W gave a 327 K peak instead of 343 K.

**Future scope:** let $k$ depend on $T$ and repeat the LP until it settles; make block positions decision variables; budget power over time using $Z_{\mathrm{th}}(s)$; validate against HotSpot.

## References

[1] F. P. Incropera, D. P. DeWitt, T. L. Bergman, and A. S. Lavine, *Fundamentals of Heat and Mass Transfer*, 7th ed. John Wiley & Sons, 2011.

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[3] W. Huang, S. Ghosh, S. Velusamy, K. Sankaranarayanan, K. Skadron, and M. R. Stan, "HotSpot: A compact thermal modeling methodology for early-stage VLSI design," *IEEE Trans. VLSI Syst.*, vol. 14, no. 5, pp. 501–513, 2006.

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